Competitive Zero Forcing: A Novel Two-Player Graph Coloring Game

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INTRODUCTION

Zero-forcing is a graph coloring process in which a colored vertex forces its only uncolored neighbor to become colored. The zero-forcing number Z(G) is the minimum number of vertices needed to color an entire graph through this process. We introduce competitive zero forcing, a two-player game built on this framework. Two players, Blue and Red, begin with disjoint minimum zero forcing sets and take turns forcing vertices. A player may also steal an opponent's vertex if it is their only non-allied neighbor and they hold a strict majority among that vertex's neighbors.

We investigate three potential sources of advantage: turn order, starting position, and choice of forcing move, examining how each manifests across cycles, paths, and spider graphs. We also explore propagation reversals, structurally opposite zero forcing sets of equal size and propagation time as natural competitors. To support our analysis, we developed a web-based Zero Forcing Set Analyzer with competitive variants and interactive play modes.

This work contributes a competitive dimension to combinatorics with applications in quantum control, information spread modeling, and bounding the minimum rank of matrices.

BACKGROUND / DEFINITIONS

Zero forcing is a graph coloring process introduced by the AIM Minimum Rank Special Graphs Work Group in 2008. Given a graph G with an initial set of colored (filled) vertices, the **zero forcing propagation rule** states: if a colored vertex v has exactly one uncolored neighbor u, then v **forces** u to become colored. The **zero forcing number** Z(G) is the minimum size of an initial colored set, called a **zero forcing set**, needed to color all vertices of G through repeated application of this rule. Zero forcing has since found applications in minimum rank problems, quantum control, and network monitoring.

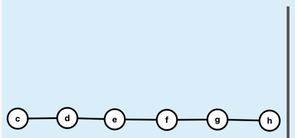
Definition 1: Zero forcing number. Let G be a graph. A subset $S \subseteq V(G)$ is a **zero forcing set** if coloring S and repeatedly applying the rule, a colored vertex with exactly one uncolored neighbor forces that neighbor to become colored, eventually colors all of $V(G)$. The **zero forcing number** $Z(G)$ is the minimum size of such a set.

Definition 2: Competitive Zero Forcing. Let $S_B, S_R \subseteq V(G)$ be disjoint minimum zero forcing sets assigned to players Blue and Red. The **competitive zero forcing game** is played either **(a) simultaneously** or **(b) sequentially**. Players may force uncolored vertices or steal opponent vertices via a strict neighbor-majority condition. The player controlling more vertices at termination wins.

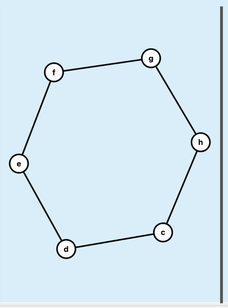
GRAPH TYPES

A graph G consists of a vertex set $V(G)$ and an edge set $E(G)$. Vertices represent the objects of interest and are drawn as circles in a diagram. Edges are connections between pairs of vertices, drawn as lines. Two vertices joined by an edge are called **adjacent** or **neighbors**. The **degree** of a vertex v, denoted $\deg(v)$, is the number of neighbors it has. A vertex of degree one is called a **leaf**, and a vertex of degree three or more is a **branch point**. A **path** P_n is a graph on n vertices connected in a single line with no cycles. A **cycle** C_n is a path whose endpoints are also adjacent, forming a closed loop. A **tree** is a connected graph with no cycles, and a **spider graph** is a tree with exactly one branch point, from which multiple paths (called **legs**) extend outward.

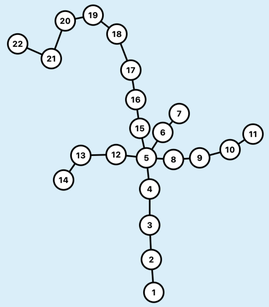
Paths



Cycles



Spiders



ZERO FORCING DEFINITIONS

Propagation: Let $B^{(i)}$ denote the set of vertices forced at time step i, and $B^{[i]} = B \cup B^{(1)} \cup \cdots \cup B^{(i)}$ the cumulative set of colored vertices after step i. The **propagation time** of B, denoted $\mathrm{pt}(G; B)$, is the least t such that $B^{[t]} = V(G)$, or infinity if B is not a zero forcing set. The **propagation time of** G is $\mathrm{pt}(G) = \min\{pt(G; B) : B \textit{ is a minimum zero forcing set}\}$.

Efficient set: A minimum zero forcing set B of G is **efficient** if it achieves the shortest propagation time among all minimum zero forcing sets. Formally, B is efficient if $|B| = Z(G)$ and $\mathrm{pt}(G; B) = pt(G)$, where $\mathrm{pt}(G) = \min\{pt(G; B) : B \textit{ is a minimum zero forcing set}\}$.

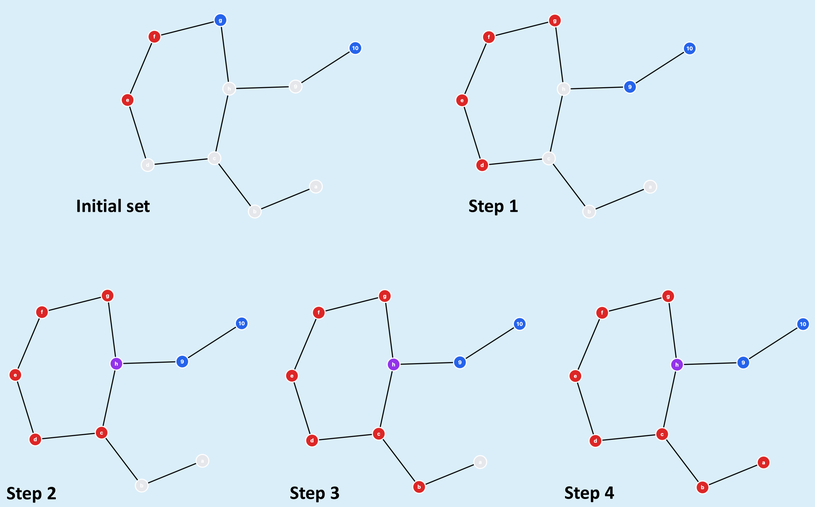
GAME 1 RULES

Each round, both Blue and Red compute all legal moves at the same time. A colored vertex v may force neighbor u if u is the **only** neighbor of v that is not the same color as v (all other neighbors must be allies). All resolutions apply together:

- If u is uncolored and only one color targets it, that color claims it.
- If both colors target the same uncolored u, count each color's neighbors of u, higher support wins; a tie makes u **purple**.
- Flipping an opponent-colored u requires a **strict majority** of the attacker's color over the defender's among u's colored neighbors.
- Purple neighbors count as support for **both** colors during resolution.

Play continues until all vertices are colored or a full round yields no changes.

GAME 1 EXAMPLE



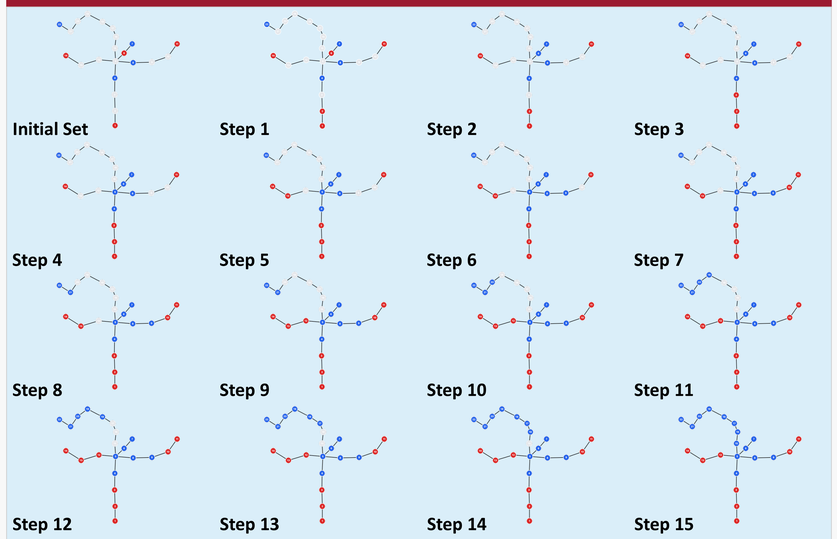
GAME 2 RULES

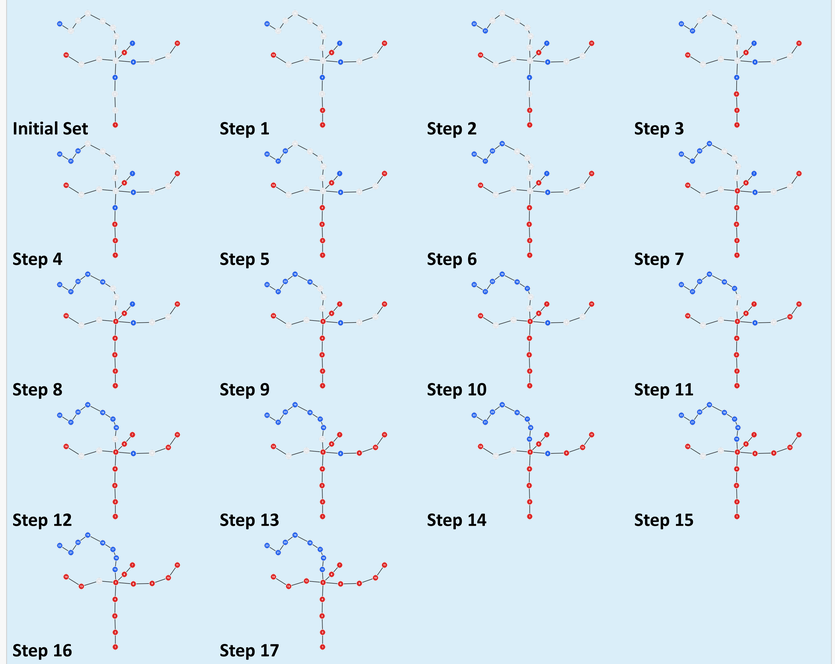
Red moves first, then Blue, alternating, once force at a time. A colored vertex v may force neighbor u if u is the **only** neighbor of v that is either uncolored or the opponent's color, every other neighbor of v must be the same color as v.

- If u is uncolored, the force **always** succeeds.
- If u is opponent-colored, the force succeeds only if the mover's color has a **strict majority** ($>$, not $\geq$) among u's currently colored neighbors.

If no legal force exists, the player passes. The game ends when all vertices are colored or **two consecutive passes** occur.

GAME 2 EXAMPLES OF CHOICES





FUTURE DIRECTIONS

- **Can Sprague-Grundy theory apply to sequential competitive zero forcing?** The sequential game is a finite two-player combinatorial game with no chance element. Assigning Sprague-Grundy values to positions would classify which graphs admit winning strategies versus forced draws.
- **Does competitive zero forcing extend to directed graphs?** Forcing along arc direction introduces asymmetric reachability that could break player symmetry even with equal-size starting sets.
- **Can simultaneous competitive zero forcing inform quantum network design?** Contested ("purple") vertices, colored by neither player exclusively, parallel qubit superposition. Investigating whether purple vertex placement informs qubit allocation in quantum control is a natural next step.

References

[1] Hogben, L. et al. Zero Forcing Sets and the Minimum Rank of Graphs. *Linear Algebra Appl.*, 2008.

[2] Butler, Steve, and Michael Young. "Throttling Zero Forcing Propagation Speed on Graphs." *Australasian Journal of Combinatorics*, vol. 57, 2013, pp. 65-71.

[3] Flagg, Mary, and Violeta Vasilevska. "Fun Graph Coloring Games and Electric Grids." *A CCICADA Homeland Security Module*, Reconnect 2017 Workshop, June 2017.

[4] Google. (2026). *Gemini* [Large language model]. https://gemini.google.com

[5] *Zero Forcing Set Analyzer* [Software]. Retrieved April 2026, from https://zeroforcingsetanalyzer.netlify.app/

[6] Anthropic. (2026). *Claude* [Large language model]. https://www.anthropic.com

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